LONGEST STAY
DETECTION.

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INTRODUCE MYSELF

WISIT SUWANNAO

Super AI Engineer

Season 6 Level 2

EXPERIENCE

- **AI Engineer Intern** at Oneput Technology Company Limited
- **Winner**, Super AI Engineer SS6 Level 2 - Demand Forecasting Coffee Chain Hackathon
- **Top 10 Finalists** - CAI CAMP 2025: PromoAutomate
- **Former participants** in Super AI Engineer Season 5 Level 1, Level 2(Online)

LinkedIn

GitHub

AGENDA

Problem Statement and Constraints

Solution of the Problem

- Definition of Stationary
- End-to-End Pipeline
- Robustness Decisions
- Result Preview & Final Answer

Deliverables and Repository

Limitations and Next Steps

PROBLEM STATEMENT AND CONSTRAINTS

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- **Input:** video raw footage from entrance.mov
- Detect the person who remained stationary the longest
- Report track_id, duration, start time และ end time
- Read FPS and duration from OpenCV metadata
- **Output:** source code, README, requirements, annotated video และ short idea description

entrance.mov
raw video footage

SOLUTION OF THE PROBLEM

SOLUTION OF THE PROBLEM

Definition of Stationary

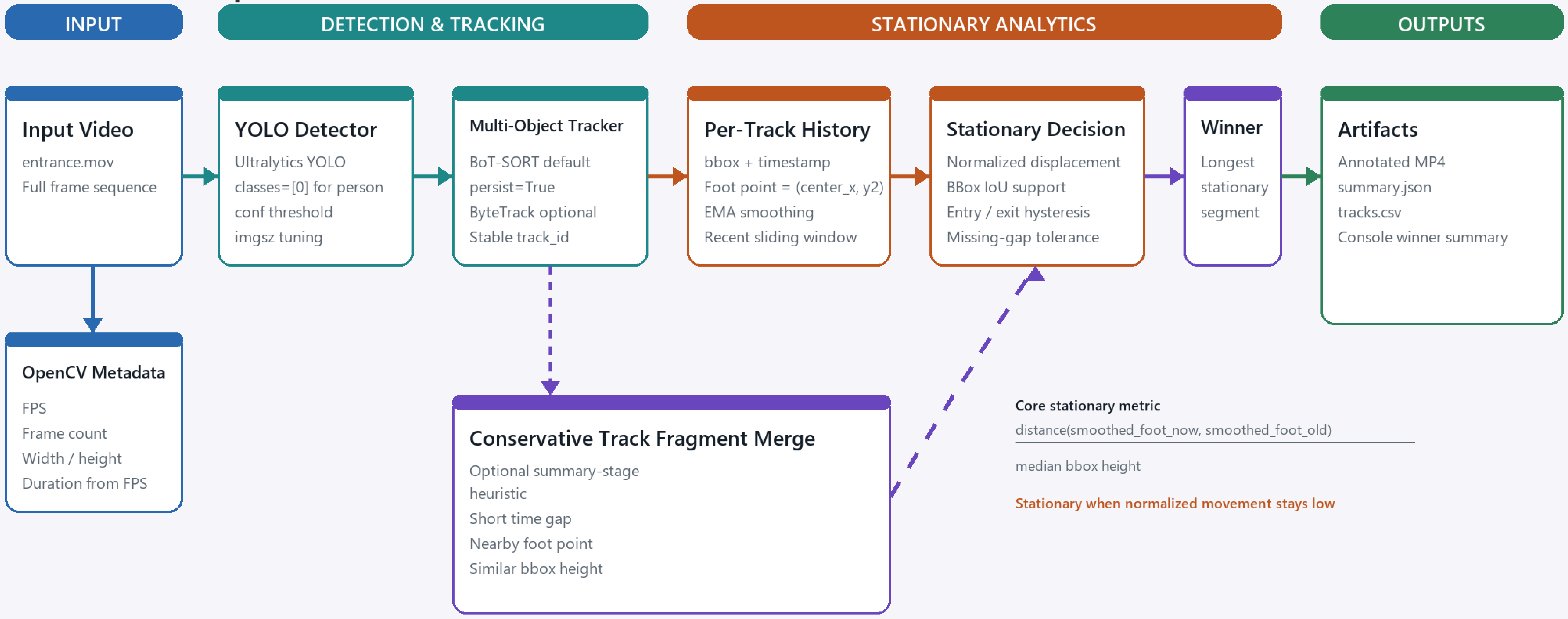
Stationary = scene position is almost unchanged

- foot_point: ใช้ (center_x, y2) แทนตำแหน่งคนบนพื้น
- norm_disp = distance(current_smoothed_foot, old_smoothed_foot) / median_bbox_height

ใช้แทนระยะการเคลื่อนที่ด้วย bbox height เพื่อให้ threshold ใช้ได้ดีขึ้นกับทั้งคนใกล้และไกลกล้องครับ

SOLUTION OF THE PROBLEM

End-to-End Pipeline



Design principle: measure each tracked person's scene position over time; do not rely on frame differencing alone.

Result: track_id 161 | longest stationary duration 41.094 seconds | interval 33.699s - 74.793s

SOLUTION OF THE PROBLEM

Robustness Decisions

1. **Detection jitter**: EMA smoothing + bbox IoU

→

2. **Short interruption**: entry/exit hysteresis + missing-gap tolerance

→

3. **Track fragmentation:** conservative merge heuristic

SOLUTION OF THE PROBLEM

Result Preview & Final Answer

1920x1080 | 29.883 FPS | 2556 frames | 85.535 seconds

Winner:
track_id 161
Longest stationary duration:
41.094 seconds
Interval:
33.699s - 74.793s

DELIVERABLES AND REPOSITORY

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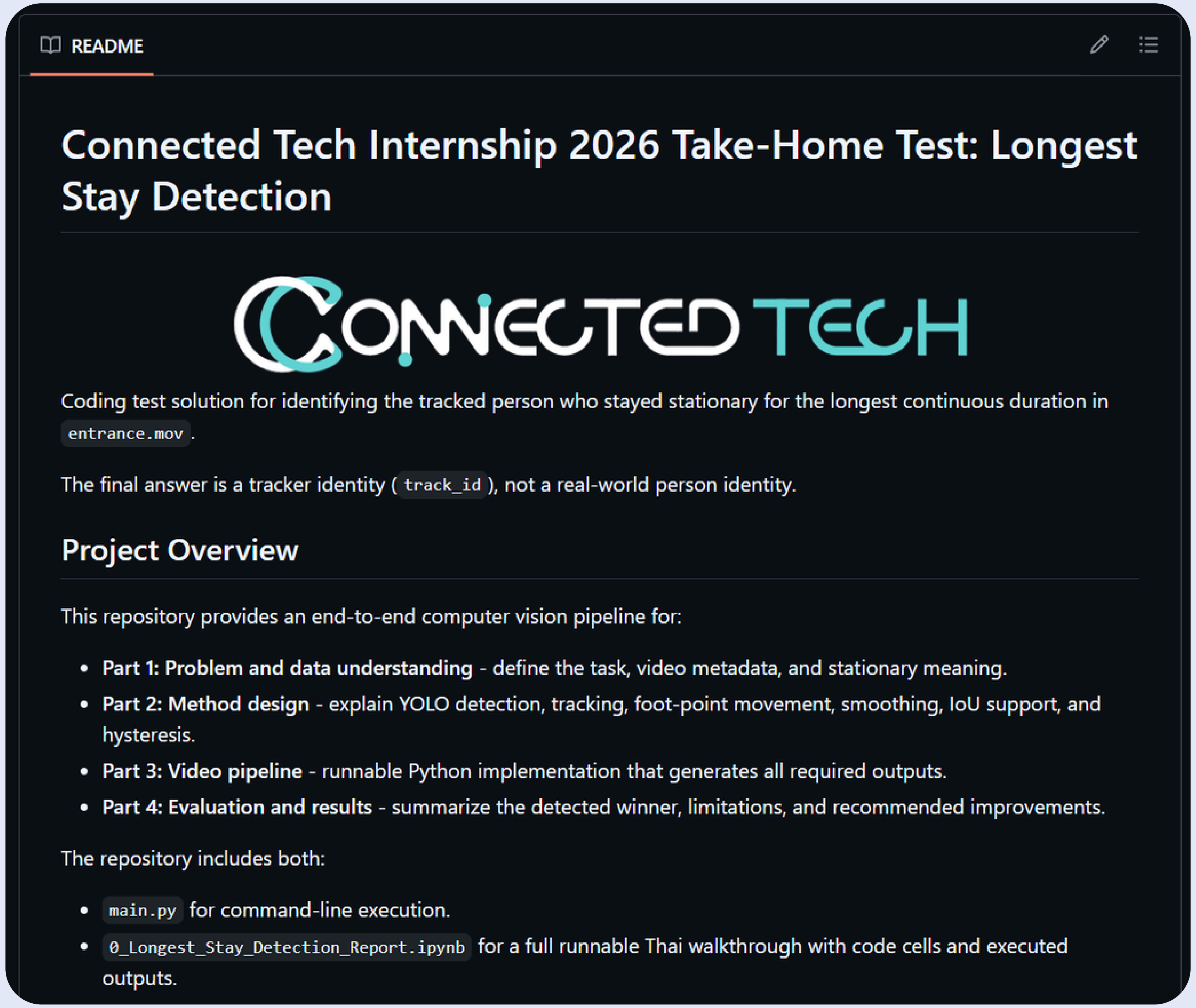
GitHub Repositories

- README.md (Example shown in the figure on the right)
- requirements.txt
- 0_Longest_Stay_Detection_Report.ipynb (Recommended for a detailed code walkthrough)
- results folder that included
  - annotated_entrance.mp4
  - summary.json
  - tracks.csv

GitHub Repository



README

Connected Tech Internship 2026 Take-Home Test: Longest Stay Detection

Coding test solution for identifying the tracked person who stayed stationary for the longest continuous duration in `entrance.mov`.

The final answer is a tracker identity (`track_id`), not a real-world person identity.

Project Overview

This repository provides an end-to-end computer vision pipeline for:

- **Part 1: Problem and data understanding** - define the task, video metadata, and stationary meaning.
- **Part 2: Method design** - explain YOLO detection, tracking, foot-point movement, smoothing, IoU support, and hysteresis.
- **Part 3: Video pipeline** - runnable Python implementation that generates all required outputs.
- **Part 4: Evaluation and results** - summarize the detected winner, limitations, and recommended improvements.

The repository includes both:

- `main.py` for command-line execution.
- `0_Longest_Stay_Detection_Report.ipynb` for a full runnable Thai walkthrough with code cells and executed outputs.

LIMITATIONS AND NEXT STEPS

LIMITATIONS AND NEXT STEPS

- ID switch during occlusion
- Detector jitter
- Perspective differences
- Camera shake
- Compare yolo11s/imgsz960 and bytetrack.yaml
- Add global motion compensation if needed

THANK YOU